

Summary of 「Time-series clustering – A decade review」

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This paper discusses the method of time-series clustering. Clustering is a data mining technique which can identify structures in an unlabeled dataset by organizing data into similar groups.

Many data have been stored in the form of time-series data, for example sales data, stock prices, weather data, biometrics data, etc. The purpose of time-series-related research includes: (1) subsequence matching, (2) anomaly detection, (3) motif discovery, (4) indexing, (5) clustering, (6) classification, (7) visualization, (8) segmentation, (9) identifying patterns, (10) trend analysis, (11) summarization, and (12) forecasting. This research work is aimed to represent an updated investigation on the trend of improvements in efficiency, quality and complexity of clustering time-series approaches during the last decade and enlighten new paths for future works.

In section 1.2, the applications of time-series clustering include:

(1) anomaly, novelty or discord detection: Discover unusual and unexpected patterns which happen in datasets surprisingly.

(2) recognizing dynamic changes in time-series: Detection of correlation between time-series. For example, in financial databases, it can be used to find the companies with similar stock price move.

(3) Prediction and recommendation: For example, in scientific databases, it can address problems such as finding the patterns of solar magnetic wind to predict today's pattern.

(4) Pattern discovery: to discover the interesting patterns in databases. For example, in marketing database, different daily patterns of sales of a specific product in a store can be discovered.

1.3 Taxonomy of time-series clustering

In section 1.3, most of clustering time-series related works are classified into three categories: (1) whole time-series clustering, (2) subsequence clustering and (3) time point clustering.

No.	Type	Content
1	Whole time-series clustering	Clustering of a set of individual time-series with respect to their similarity. Here, clustering means applying conventional clustering on discrete objects, where objects are time-series.
2	Subsequence clustering	clustering on a set of subsequences of a time-series that are extracted via a sliding window, that is, clustering of segments from a single long time-series.
3	Time point clustering	clustering of time points based on a combination of their temporal proximity of time points and the similarity of the corresponding values. This approach is similar to time-series segmentation. However, it is different from segmentation as all points do not need to be assigned to clusters, i.e., some of them are considered as noise.

2. Representation methods for time series clustering

No.	Approach	Content
1	Data adaptive	Performed on all time-series in datasets and try to minimize the global reconstruction error using arbitrary length (non-equal) segments.
2	Non-data adaptive	Suitable for time-series with fix size (equal-length) segmentation, and the comparison of representations of several time-series is straightforward. The methods in this group are Wavelets: HAAR, DAUBECHIES, Coeiflets, Symlets, Discrete Wavelet Transform(DWT), spectral Chebyshev Polynomials, spectralDFT, Random Mappings, Piecewise Aggregate Approximation (PAA) and Indexable Piecewise Linear Approximation (IPLA)
3	Model based	represent a time-series in a stochastic way such as Markov Models and Hidden Markov Model (HMM), Statistical Models, Time-series

		Bitmaps, and Auto-Regressive Moving Average (ARMA). User can define the compression-ratio based on the application in hand.
4	Data dictated	the compression ratio is defined automatically based on raw time-series such as Clipped.

3. Similarity/dissimilarity measures in time-series clustering

No.	Find similar time-series in	Content
1	Time	Fourier transforms, wavelets or Piecewise Aggregate Approximation (PAA).
2	Shape	Dynamic time Warping (DTW)
3	change (structural similarity)	Hidden Markov Models (HMM) or an ARMA process

No.	Level	Content
1	Shape	measure similarity in short-length time-series clustering such as expression profiles or individual heartbeats by comparing their local patterns. Shape-based similarity measure is to find the similar time-series in time and shape, such as Euclidean, DTW, LCSS, MVM. It is a group of methods which are proper for short time-series.
2	Structure	measures similarity which is based on global and high-level structure, and it is used for long-length time-series data.

No.	Type of distance measure	Content
1	Shape-based	Euclidean, DTW, LCSS, MVM. It is a group of methods which are proper for short time-series.
2	Compression based	Suitable for short and long time-series, such as CDM, Autocorrelation, Short time series distance, Pearson's correlation coefficient and related distances, Cepstrum, Piecewise normalization and Cosine wavelets.
3	Feature based	proper for long time-series, such as Statistics, Coefficients

4	Model based	proper for long time-series, such as HMM and ARMA.
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4. Time-series cluster prototypes

No.	Prototype used	Content
1	medoid	1. Euclidean 2. Dynamic Time Warping (DTW)
2	average	1. Shape averaging using Dynamic Time Warping (DTW) 2. Shape averaging using Longest Common Sub-Sequence (LCSS)
3	local search	Hautamaki et al. They apply medoid, average and local search on k-Medoids, Random Swap (RS) and Agglomerative Hierarchical clustering (where k-means is used to fine-tune the output) to evaluate their work. They figured out that local search provides the best clustering accuracy and also more improvement to k-Medoids.

5. Time-series clustering algorithms

No.	Clustering algorithms	Content
1	Hierarchical	weak in terms of quality, usually hierarchical clustering algorithms are combined with another algorithm as a hybrid clustering approach to remedy this issue.
2	Partitioning	k-Means, k-Medoids (PAM), FCM (Fuzzy c-Means), Fuzzy c-Medoids algorithm
3	Model-based	COBWEB, or Neural Network approaches, e.g., ART or Self-Organization Map (SOM), polynomial models, Gaussian mixed models, ARIMA, Markov chain and Hidden Markov models.
4	Density-based	DBSCAN, OPTICS
5	Grid-based	STING and Wave Cluster
6	Multi-step	3-PhaseTime series Clustering model (3PTC), Two-step Time series Clustering (TTC) algorithm

6. Time-series clustering evaluation measures

No.		Content
1	External index	1. Cluster purity 2. Cluster Similarity Measure (CSM) 3. Folkes and Mallow index (FM) 4. Jaccard Score 5. Rand index (RI) 6. Adjusted Rand Index (ARI) 7. F-measure 8. Normalized Mutual Information (NMI) 9. Entropy
2	Internal index	1. SSE